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INTOXICOM Workshop Report: Making toxicology tools more accessible and interoperable

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Introduction

As part of the INTOXICOM Implementation Study for the ELIXIR Toxicology Community a series of workshops is organized (Martens et al., 2024). Here, we here report on the 3rd workshop, titled "Making toxicology tools more accessible and interoperable" which was held from 26 to 27 March 2025 at the SciLifeLab at Uppsala University in Sweden.

The workshop welcomed 29 participants from Sweden, The Netherlands, Cyprus, Switzerland, Italy, Greece, France, and Norway. This 3rd INTOXICOM workshop covered various aspects of making computational tools available and how they are used. Several projects, such as NanoSolveIT (Afantitis et al., 2020), ONTOX (Vinken et al., 2021), and VHP4Safety (Kienhuis et al., 2024), already make computational toxicology services available, but the field of integrating computional toxicology dates back much longer, such as Bioclipse developed at Uppsala University (Willighagen et al., 2011), making it a perfect location to have held this workshop.

Presentations

Organizers and invited speakers held several presentation to provide information to support the workshop, scheduled as part of sessions hosted by Ola Spjuth and Egon Willighagen:

Speaker	Talk Title
Egon Willighagen	ELIXIR Introduction
Jonne Rietdijk	User perspective: Morphological Profiling for Chemical Toxicity Assessment
Ivo Djidrovski	Large Language Model tools and AI-Agents in toxicology research
Antreas Afantitis	NovaMechanics platforms: NanoSolveIT, Enalos Cloud, Pharos DMS
Ziye Zheng	Toxicity prediction with regulatory relevance: predicting the CLP H-statement
Egon Willighagen	Toxicity prediction with regulatory relevance: predicting the CLP H-statement (Willighagen, 2025)

Speaker	Talk Title
Nikita Churikov	Sharing and serving ML models in toxicology research with SciLifeLab Serve
Jente Houweling	VHP4Safety project for data/tool integration

Results

Day 1

After a *Welcome at SciLifeLab* by Ola Spjuth and an overview of ELIXIR Europe, the INTOXICOM workshop, and the schedule of the workshop, the first session what about *What tools for toxicology exist?*. The focus of this session was aimed at relevent services indexed by the ELIXIR Toxicology Community on bio.tools for computational tools and FAIRsharing for databases:

- bio.tools records annotated with toxicology
- FAIRsharing's toxicology collection

The participants that several services were missing in the collection of 103 tools found in bio.tools, including VHP4Safety services, TXG-MAPr, BMDExpress, the Enalos Cloud Platform, OPERA and VEGA, EFSA tools, and https://www.parc-models.eu/tools/.

The FAIRsharing collection was already discussed at the second INTOXICOM workshop (Bonatto Minella et al., 2026) and futher curated by Bonatto Minella *et al.* (Bonatto Minella et al., 2025). During the workshop several databases were found to be missing, including ToxVal, ToxRef, ToxCast, AOP-Wiki, ECHA CHEM, Nanosafety Data Interface, GenRA, chemPharos, RepDose Database Fraunhofer ITEM, and Le Portail Substances Chimiques (PSC).

In the afternoon, the second session consisted of three user stories on data and tools in the session *How to make your data integrate with tools?*, chaired by Ola. The first story was given by Jonne Rietdijk on morphological profiling for chemical toxicity, proposed for toxicology testing in 2010 (Krewski et al., 2010). Riedijk walked the participants through various approaches, such as cell painting. Data was shared on Figshare (Rietdijk et al., 2022), comparing effects of exposure to cetyltrimethylammonium bromide, bisphenol A, and bibutyltin dilaurate in several human cell lines. Images can be shared with OMERO (Allan et al., 2012), which has advantages over the more general Figshare.

The story provided by Ivo Djidrovski describes the rapid progress of large language models (LLMs). His work focuses on using LLM agents capabilities, which provide access to data and knowledge bases. He demoes the use of LLMs in a chemical hazard assistant, based on the OECD QSAR Toolbox (Djidrovski et al., 2026).

After a break, Antreas Afantitis descrbes the NovaMechanics platforms NanoSolveIT, Enalos Cloud, Pharos DMS, based on 20 years of experience in the field. Enalos cloud enables non-expert users to leverage ML. Model outputs are accompanied by a workflow of what the user has performed. The Nanopharos database integrates experimental data with physics-based models and machine learning.

The last user story was provided by Ziye Zheng fo the IVL Swedish Environmental Institute. It discussed SafeChem, the regulatory relevance, lack of negative controls, data mining from the ECHA REACH dossiers, and computational toxicology approaches, including molecular descriptors, fingerprints, and descriptor-free approaches, and comparing traditional machine learning and deep learning.

During the discussion at the end of this session, the workshop participants identified several missing things (some gaps are filled by the wider community in the year following the workshop): model context protocols (MCPs) for toxicology, Open API definitions, clarity and/in documentation, also for knowledge graphs, and better interoperability standards, for example, with FAIR Implementation Profiles.

Day 2

The second day started with the session *How can solutions complement your data? Using tools in tox studies* with three presentations and an open discussion. The first presentation was given by Egon Willighagen on how knowledge bases give context to experimental data (Willighagen, 2025). For toxicology, these may include WikiPathways (Agrawal et al., 2023), and AOP-Wiki (Martens et al., 2022), e.g. via the AOP-Wiki RDF. The importance of knowledge bases is that they curate the collected scientific knowledge. Bio2RDF gets a shout out for making knowledge bases interoperable (Belleau et al., 2008).

The second talk was given by Nikita Churikov of SciLifeLab on *Sharing and serving ML models in toxicology research with SciLifeLab Serve*. Serve is an infrastructure for bioinformaticians to publish open source codes (see Figure 1). The platform uses Docker to make tools available, such as DICTrank Predictor (Seal et al., 2024). The talk further discussed exchange of metadata and minimal reporting standards for tools and the Five recommandations for FAIR software and the Self-assessment for FAIR research software come up.

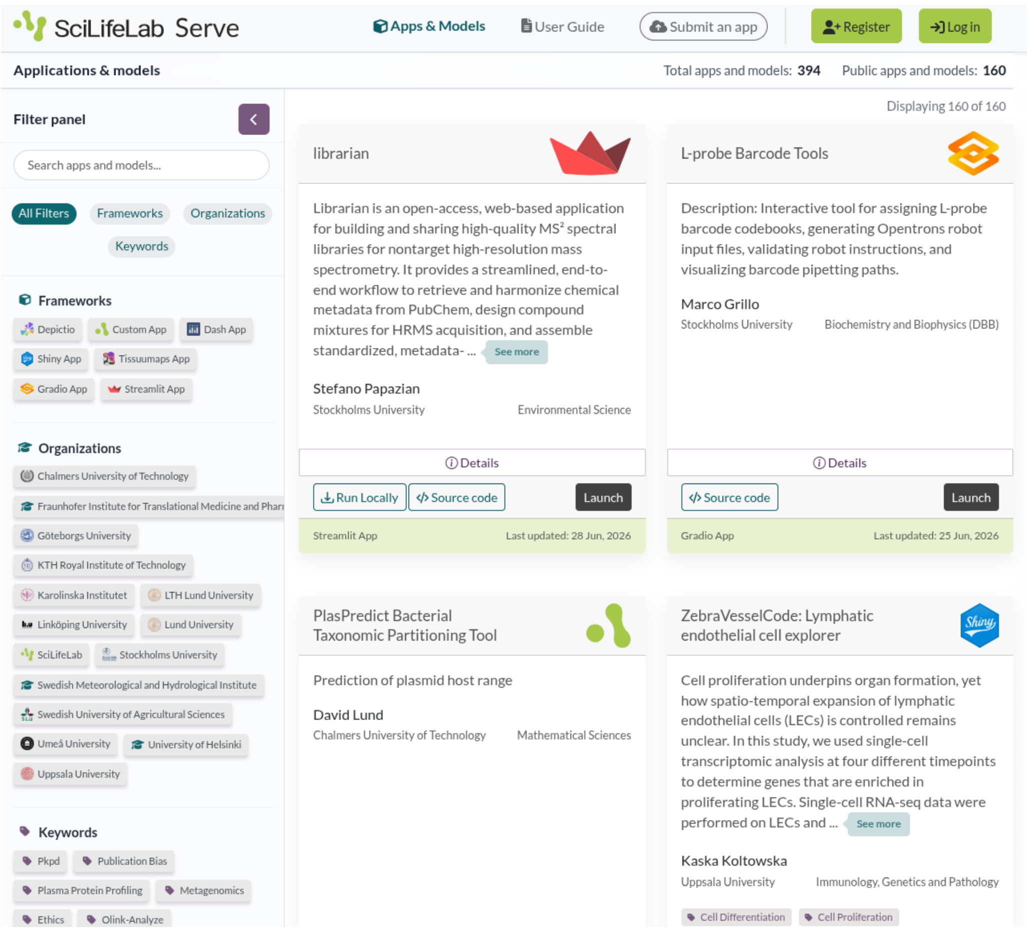

Figure 1: Screenshot of the SciLifeLab Serve website.

The last talk in this session was given my Jente Houweling working at the Dutch National Institute for Public Health and the Environment. She introduces the participants to the

VHP4Safety project and her research on the ToxTempAssistant (Houweling et al., 2026) in particular. This tool uses large language models to support filling out ToxTemp templates, aimed at providing sufficient details on experimental toxicology data (Krebs et al., 2019).

Discussion

All sessions has many lively discussions with a lot more pointers and actions. Of particularly interest is the interest in using the interoperability that toxicologists have worked on over the years to enable LLMs to interface with knowledge bases and tools. This interest is shared by many, and ELIXIR Europe now has an AI Ecosystem Focus Group, which includes two task forces, one on agentic AI and the other on generative AI.

Another important point is that finding tools and knowledge bases is a continous activity, and the Toxicology Community will continue have to be active in ensuring they are easily found (as part of FAIR). During the workshop several tools and databases were identified which were not found in bio.tools of FAIRsharing at the time, and they need to be added. Platforms like VHP4Safety, NanoSolveIT, and SciLifeLab Serve should make metadata interoperable, e.g. with BioSchemas.

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References

Afantitis, A., Melagraki, G., Isigonis, P., Tsoumanis, A., Varsou, D. D., Valsami-Jones, E., Papadiamantis, A., Ellis, L.-J. A., Sarimveis, H., Doganis, P., Karatzas, P., Tsiros, P., Liampa, I., Lobaskin, V., Greco, D., Serra, A., Kinaret, P. A. S., Saarimäki, L. A., Grafström, R., ... Lynch, I. (2020). NanoSolveIT Project: Driving nanoinformatics research to develop innovative and integrated tools for in silico nanosafety assessment. *Computational and Structural Biotechnology Journal*, *18*, 583–602. https://doi.org/10.1016/J.CSBJ.2020.02.023 **[cito:obtainsBackgroundFrom]**

Agrawal, A., Balcı, H., Hanspers, K., Coort, S. L., Martens, M., Slenter, D. N., Ehrhart, F., Digles, D., Waagmeester, A., Wassink, I., Abbassi-Daloii, T., Lopes, E. N., Iyer, A., Acosta, J. M., Willighagen, L. G., Nishida, K., Riutta, A., Basaric, H., Evelo, C. T., ... Pico, A. R. (2023). WikiPathways 2024: Next generation pathway database. *Nucleic Acids Research*. https://doi.org/10.1093/NAR/GKAD960 **[cito:citesForInformation]**

Allan, C., Burel, J.-M., Moore, J., Blackburn, C., Linkert, M., Loynton, S., Macdonald, D., Moore, W. J., Neves, C., Patterson, A., Porter, M., Tarkowska, A., Loranger, B., Avondo, J., Lagerstedt, I., Lianas, L., Leo, S., Hands, K., Hay, R. T., ... Swedlow, J. R. (2012). OMERO: Flexible, model-driven data management for experimental biology. *Nature Methods*, *9*(3), 245–253. https://doi.org/10.1038/NMETH.1896 **[cito:citesForInformation]**

Belleau, F., Nolin, M.-A., Tourigny, N., Rigault, P., & Morissette, J. (2008). Bio2RDF: Towards a mashup to build bioinformatics knowledge systems. *Journal of Biomedical Informatics*, *41*(5), 706–716. https://doi.org/10.1016/J.JBI.2008.03.004 **[cito:credits]**

Bonatto Minella, C., Bünger, M., Durník, R., Exner, T., Gamba, A., Houweling, J., Ioannidis, V., Jennen, D., Guerrero, L. A. R., Lynch, I., Maran, U., Komanec, K., Stierum, R., Martens, M., Morsy, S., & Willighagen, E. (2026). INTOXICOM Workshop Report: Advancing FAIR toxicology research output across Europe: ELIXIR & FAIRsharing studies. *BioHackrXiv*. https://doi.org/10.37044/OSF.IO/64DFV_V1 **[cito:citesForInformation]**

Bonatto Minella, C., Willighagen, E., Martens, M., Lister, A., & Granell, R. (2025). *The ELIXIR toxicology community collection of FAIR-enabling standards and data resources*. Zenodo. https://doi.org/10.5281/zenodo.15799783 **[cito:citesForInformation]**

Djidrovski, I., Pieters, R., Legler, J., & Teunis, M. (2026). O-QT assistant: A multi-agent AI system for streamlined chemical hazard assessment and read-across analysis using the OECD QSAR toolbox API. *Computational Toxicology*, *37*, 100395. https://doi.org/10.1016/j.comtox.2025.100395 **[cito:citesAsRecommendedReading]**

Houweling, J. M., Arras, M. M., Willighagen, E. L., Jennen, D. G., Evelo, C. T., & Kienhuis, A. S. (2026). ToxTempAssistant: Using large language models to standardise cell-based toxicological test method descriptions. *Evidence-Based Toxicology*, *4*(1), 2638036. **[cito:citesForInformation]**

Kienhuis, A., Krul, C., Engelen, J. van, Evelo, C. T., Hessel, E., Hoekman, J., Kramer, N., Krop, E., Masereeuw, R., Moors, E., O. Negro, S., Piersma, A. H., Pieters, R., Teunis, M., Willighagen, E. L., & Legler, J. (2024). The Virtual Human Platform for Safety Assessment (VHP4Safety) project: Next generation chemical safety assessment based on human data. *ALTEX*. https://doi.org/10.14573/ALTEX.2407211 **[cito:obtainsBackgroundFrom]**

Krebs, A., Waldmann, T., Wilks, M. F., Van Vugt-Lussenburg, B. M. A., Van der Burg, B., Terron, A., Steger-Hartmann, T., Ruegg, J., Rovida, C., Pedersen, E., Pallocca, G., Luijten, M., Leite, S. B., Kustermann, S., Kamp, H., Hoeng, J., Hewitt, P., Herzler, M., Hengstler, J. G., … Leist, M. (2019). Template for the description of cell-based toxicological test methods to allow evaluation and regulatory use of the data. *ALTEX*, *36*(4), 682–699. https://doi.org/10.14573/ALTEX.1909271 **[cito:citesForInformation]**

Krewski, D., Acosta, D., Andersen, M., Anderson, H., Bailar, J. C., Boekelheide, K., Brent, R., Charnley, G., Cheung, V. G., Green, S., Kelsey, K. T., Kerkvliet, N. I., Li, A. A., McCray, L., Meyer, O., Patterson, R. D., Pennie, W., Scala, R. A., Solomon, G. M., … Zeise, L. (2010). Toxicity Testing in the 21st Century: A Vision and a Strategy. *Journal of Toxicology and Environmental Health – Part B – Critical Reviews*, *13*(2-4), 51–138. https://doi.org/10.1080/10937404.2010.483176 **[cito:discusses]**

Martens, M., Evelo, C. T., & Willighagen, E. L. (2022). Providing Adverse Outcome Pathways from the AOP-Wiki in a Semantic Web Format to Increase Usability and Accessibility of the Content. *Applied In Vitro Toxicology*. https://doi.org/10.1089/AIVT.2021.0010 **[cito:citesForInformation]**

Martens, M., Lynch, I., Exner, T. E., Stierum, R., Nymark, P., Martinát, D., Acosta, J. M., Wittig, U., Wittwehr, C., Bünger, M., Burger, G., Abbassi-Daloii, T., Killi, U., Houweling, J., Chowdhury, R. R., Danilyuk, T., Sinke, L., Çınar, O., & Willighagen, E. (2024). INTOXICOM Workshop Report: FAIRification of Toxicological Research Output: Leveraging ELIXIR Resources. *BioHackrXiv*. https://doi.org/10.37044/OSF.IO/UN2RW **[cito:citesAsRecommendedReading] [cito:citesAsEvidence]**

Rietdijk, J., Aggarwal, T., Georgiev, P., Lapins, M., Carreras-Puigvert, J., & Spjuth, O. (2022). *Morphological profiling of environmental chemical combinations - images, pipelines, and features*. https://doi.org/10.17044/scilifelab.19063028.v1 **[cito:discusses]**

Seal, S., Spjuth, O., Hosseini-Gerami, L., García-Ortegón, M., Singh, S., Bender, A., & Carpenter, A. E. (2024). Insights into Drug Cardiotoxicity from Biological and Chemical Data: The First Public Classifiers for FDA Drug-Induced Cardiotoxicity Rank. *Journal of Chemical Information and Modeling*, *64*(4), 1172–1186. https://doi.org/10.1021/ACS.JCIM.3C01834 **[cito:citesAsRelated]**

Vinken, M., Benfenati, E., Busquet, F., Castell, J., Clevert, D.-A., Kok, T. M. de, Dirven, H., Fritsche, E., Geris, L., Gozalbes, R., Hartung, T., Jennen, D., Jover, R., Kandarova, H., Kramer, N., Krul, C., Luechtefeld, T., Masereeuw, R., Roggen, E., … Piersma, A. H. (2021).

Safer chemicals using less animals: Kick-off of the European ONTOX project. *458*, 152846. https://doi.org/10.1016/J.TOX.2021.152846 **[cito:obtainsBackgroundFrom]**

Willighagen, E. (2025). *Giving biological and toxicological context: WikiPathways, AOP-wiki, other databases*. Zenodo. https://doi.org/10.5281/zenodo.21224616 **[cito:citesAsEvidence]**

Willighagen, E., Jeliazkova, N., Hardy, B., Grafström, R., & Spjuth, O. (2011). Computational toxicology using the OpenTox application programming interface and Bioclipse. *BMC Research Notes*, *4*(1), 487. https://doi.org/10.1186/1756-0500-4-487 **[cito:obtainsBackgroundFrom]**

Author contributions

Egon Willighagen: Writing – review & editing